Non-Linear Motion from Two Frames: Cubic Hermite Trajectories for Video Frame Interpolation

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Abstract

*Video frame interpolation from two frames almost always assumes a pixel travels in a straight line between them. Methods that drop this assumption pay for it with extra frames, point tracks, or events, or they model time implicitly and run a network again at every timestep. We present **HermiteVFI**, which describes motion by the instantaneous velocity at each keyframe instead of the average displacement between them. A Hermite curve is the natural fit here: its coefficients are positions and velocities, so the two keyframes enter as fixed endpoints while the two velocities stay free to bend the path between them. Neither velocity is measured; both are latent variables the reconstruction loss learns from the frame pair. Whatever it predicts, the curve still passes through both keyframes exactly, linear and quadratic motion remain special cases, and the flow at any t is a polynomial evaluation rather than a forward pass. Velocity has to come from appearance, because a flow field is path independent and carries no curvature at all; we show which cues carry the signal and how much each one contributes. We train on X4K1000FPS for its fast non-uniform motion and evaluate on X4K1000FPS, SNU-FILM, and Vimeo90K-Septuplet. The gains are largest under difficult motion and at high interpolation ratios, where interpolating K frames costs one motion pass instead of K.*

1. Introduction

Video frame interpolation synthesizes one or more frames between two consecutive keyframes. It supports slow-motion capture, frame-rate conversion for high-refresh displays, temporal upsampling in animation pipelines, and video compression [3, 10]. The dominant family is flow based: an optical flow network estimates correspondence between the two keyframes, that motion is resampled to the query time, and a synthesis network warps and blends the two frames into the result [9, 10, 12, 17, 28]. Progress in this family has come largely from better flow estimators [19, 23] and better synthesis [16, 17], while the step in between has stayed almost unchanged.

That step is where the problem lies. Optical flow reports the total displacement between the two keyframes, and two-frame methods obtain the intermediate motion by scaling it by the query time. This assumes constant velocity along a straight line, which fails whenever objects accelerate, reverse direction, or the camera starts and stops [14, 26]. The resulting error grows with both the speed of the motion and the interpolation ratio, and training on non-uniform motion under this assumption also makes the supervision inconsistent, since the same pair and query time can correspond to different true positions [20].

Existing non-linear methods each pay for curvature somewhere. The larger group widens the temporal window: quadratic methods fit a parabola through four frames [15, 26], later work regresses multi-stage quadratic motion or switches between motion models over flows from several frames [4, 14], spline methods fit trajectories to point tracks [1], and event-based methods recover continuous motion from a second sensor [13, 24, 25]. But four frames are not always available, tracking adds an auxiliary model, and event cameras are not commodity hardware. A second group keeps two frames and models curvature implicitly, through an acceleration branch supervised by a privileged teacher [6], a coordinate network mapping query time and a motion latent to flow [5], or a bidirectional motion field [20]. Here the trajectory is never written down as a curve, so nothing guarantees that flows predicted at nearby times belong to one consistent path or agree with the keyframes they came from, and the motion model runs once per query time.

The gap is an explicit trajectory recovered from two frames alone. We close it by predicting the instantaneous velocity at each keyframe instead of the average displacement between them, and letting a cubic Hermite curve carry the motion in between. Hermite is the right carrier because its coefficients are positions and velocities: the keyframes are fixed endpoints and the velocities bend the path, so endpoint consistency is structural rather than learned, and linear and quadratic motion are special cases. The curve is analytic, so the query time costs a polynomial evaluation rather than a network pass.

The velocities cannot come from flow, which records only where a pixel started and finished. We instead treat them as latents learned by reconstruction from appearance features and a parameter-free blur descriptor, and measure each input's contribution by ablation. The parameterization also dictates the training setup: a single query time confounds the two velocities, which we argue explains a prior negative result on triplet data [6]. We therefore supervise several times per pair, training on X4K1000FPS [22] and evaluating there and on SNU-FILM [2] and Vimeo90K-Septuplet [27].

Our contributions are as follows:

- We propose to describe intermediate motion by the instantaneous velocity at each keyframe rather than the average displacement between them, which raises the order of motion the model can express without increasing the number of input frames.
- We show that the two endpoint velocities determine a cubic Hermite trajectory that passes through both keyframes for any network output, with linear and quadratic motion recovered as special cases of the same closed form, and that carries no parameters, so interpolating K frames costs one motion pass rather than K.
- We recover the endpoint velocities from two RGB frames alone, without extra frames, point tracks, or events, by treating them as latents learned from appearance features and a parameter-free blur descriptor, and we isolate the contribution of each input by ablation.
- We identify why a single query time confounds the two endpoint velocities, reinterpret a prior negative result on triplet data in these terms, and supervise multiple times per frame pair to resolve the ambiguity.

2. Related Work

Flow-based frame interpolation. Within the flow-based pipeline, methods differ mainly in how motion is resampled and how frames are composited. SuperSloMo [10] approximates the intermediate flow as a bilateral combination of the two keyframe flows, SoftSplat [17] replaces backward warping with softmax splatting to resolve many-to-one collisions, and RIFE [9] predicts the intermediate flow directly and drops flow reversal. Work aimed at multi-frame output reduces the per-frame cost: XVFI [22] predicts flow at an arbitrary query time from a scale-invariant pyramid, and M2M [7] estimates motion once per pair and splats to each target time. Both still produce motion for one time at a time, whereas our basis is fixed and parameter-free, so an extra query time costs a polynomial evaluation.

Non-linear motion. Methods that widen the temporal window differ in how the curve is fit: QVI [26] assumes constant acceleration over four frames, EQVI [15] rectifies the fit by least squares, and later work regresses multi-stage quadratic motion or selects among motion models [4, 14]. Spline methods fit trajectories to tracked points [1], and event-based methods integrate a second sensor stream [13, 24, 25]. Two-frame methods instead model curvature implicitly: IQ-VFI [6] uses a distilled acceleration prior, GIMM [5] a coordinate network conditioned on a motion latent, and BiM-VFI [20] a bidirectional motion field with a content-aware timestep. InterpAny-Clearer [31] takes a third route and replaces the scalar t with a distance map, moving the ambiguity into the input rather than resolving it.

Motion cues in appearance. Motion blur records where a pixel travelled during the exposure. Prior work reconstructs a short video from a single blurred image [11], jointly deblurs and interpolates from blurry low-frame-rate input [18, 21], and renders arbitrary intermediate times by decoding the temporal correlation encoded in blur [30]. These treat blur as a degradation to be inverted; we use it in the opposite direction, as an appearance cue from which the endpoint velocities are predicted.

Supervision for non-linear motion. Prior two-frame methods lean on privileged supervision: RIFE [9] distills intermediate flow from a teacher that receives the ground-truth frame, IQ-VFI [6] adds acceleration and motion distillation losses against a triplet teacher, and BiM-VFI [20] supervises its motion field with a knowledge-distillation term. IQ-VFI also reports that an acceleration branch trained by reconstruction alone gives no gain. That experiment uses VTF [27], i.e. a single query time, where the two endpoint velocities enter the trajectory only through one scalar combination and are not separately identifiable; we read it as a statement about identifiability rather than about photometric supervision.

3. Method

Given adjacent frames $I_0, I_1 \in \mathbb{R}^{H\times W\times 3}$, flow-based VFI estimates the motion between them, resamples it to a query time $t \in (0,1)$, and synthesizes $\hat{I}_t$ by warping and blending. We focus on the resampling step. Let $F = f_{0\to1}$ and $F' = f_{1\to0}$ be bidirectional flows from a pre-trained estimator, $\Phi(t)$ the displacement of a pixel from its position in I_0 at time t, and $\overleftarrow{\omega}$, $\overrightarrow{\omega}$ the backward warping and forward splatting operators. Primes denote quantities defined on the pixel grid of I_1. Linear motion models take $\Phi(t) = tF$.

3.1. Hermite motion representation

Endpoint velocities. F is the average velocity over $[0,1]$, not the velocity at either keyframe. We parameterize motion instead by the instantaneous endpoint velocities $m_0 = \dot{\Phi}(0)$ and $m_1 = \dot{\Phi}(1)$, which together with the endpoint positions $\Phi(0) = \mathbf{0}$ and $\Phi(1) = F$ determine a unique cubic Hermite curve,

$$\Phi(t) = (t^3 - 2t^2 + t)\, m_0 + (-2t^3 + 3t^2)\, F + (t^3 - t^2)\, m_1. \quad (1)$$

Residual form. Since m_0 and m_1 are dominated by F, we predict residuals $\delta_i = m_i - F$ rather than the velocities themselves. Substituting into Eq. (1) and collecting terms gives

$$\Phi(t) = t\,F + \beta_2(t)\,A + \beta_3(t)\,B, \quad (2)$$

with $\beta_2(s) = s^2 - s$, $\beta_3(s) = s^3 - s$, and

$$A = -2\delta_0 - \delta_1, \qquad B = \delta_0 + \delta_1. \quad (3)$$

Eq. (2) is an exact rewriting of Eq. (1): the linear model is the leading term and the network predicts only the deviation from it. Evaluating a timestep costs three tensor products and no network pass. *The full expansion is given in Sec. A.1 of the supplementary.*

Properties. Both basis functions vanish at $s = 0$ and $s = 1$, so $\Phi(0) = \mathbf{0}$ and $\Phi(1) = F$ hold for any A, B, and differentiating recovers $\dot{\Phi}(0) = F – A – B = m_0$ and $\dot{\Phi}(1) = F + A + 2B = m_1$. Setting $\delta_0 = \delta_1 = 0$ recovers linear motion exactly; $\delta_1 = -\delta_0$ gives $B = 0$ and reduces Eq. (2) to the quadratic model of [6] with $A = a_\tau/2$; appending $\beta_4(s) = s^4 - s$ extends the family to quartic.

Backward trajectory. The same construction on the grid of I_1 with time reversed gives

$$\Phi'(t) = (1-t)F' + \beta_2(1-t)A' + \beta_3(1-t)B', \quad (4)$$

with A', B' formed from δ'_0, δ'_1 as in Eq. (3).

Identifiability. At a fixed t the model exposes only $\beta_2(t)A + \beta_3(t)B$. As A and B are independent linear functions of δ_0, δ_1, a single timestep cannot separate them, and the endpoint velocities are unidentifiable from a frame triplet. We therefore supervise several timesteps per frame pair, and train on X4K1000FPS [22] and Vimeo90K-Septuplet [27], which provide eight and seven consecutive frames respectively.

3.2. Endpoint velocity estimation

Inputs. We run a frozen RAFT [23] in both directions, retaining its context features c_i, final hidden state h_i, and convex upsampling mask W_i in addition to the flows. From the flows we compute a forward–backward inconsistency map and a photometric residual,

$$U = \|F + \overleftarrow{\omega}(F', F)\|_1, \qquad Z = -\|I_0 - \overleftarrow{\omega}(I_1, F)\|_1, \quad (5)$$

used as an occlusion proxy and as splatting importance respectively.

Blur descriptor. Optical flow is path independent and carries no curvature; directional blur is the cue in a static pair that integrates motion over the exposure. From the smoothed structure tensor $\mathcal{G}_{5\times5} * (\nabla I_i \nabla I_i^\top)$, with eigenvalues $\lambda_1 \geq \lambda_2$ and dominant orientation θ, we form a parameter-free descriptor

$$\mathcal{B}_i = \big[\, \lambda_1/(\lambda_2 + \epsilon),\ \cos 2\theta,\ \sin 2\theta \,\big]. \quad (6)$$

The closed form of the eigenvalues and double-angle terms is given in Sec. A.3 of the supplementary.

Velocity head. An appearance encoder extracts features from the frame pair and blur descriptors, warped onto the grid of I_0,

$$\mathcal{S}_0 = \mathrm{AppNet}\big([\, I_0,\ \overleftarrow{\omega}(I_1, F),\ \mathcal{B}_0,\ \overleftarrow{\omega}(\mathcal{B}_1, F) \,]\big), \quad (7)$$

which are concatenated with the flow-derived inputs at $1/8$ resolution,

$$\Psi_0 = \big[\, c_0,\ h_0,\ \tfrac{1}{8}F^{\downarrow 8},\ \tfrac{1}{8}\overleftarrow{\omega}(F', F)^{\downarrow 8},\ U^{\downarrow 8},\ \mathcal{S}_0 \,\big], \quad (8)$$

and passed to a single-scale convolutional head $\mathcal{C}$ predicting both residuals,

$$[\delta_0^{\downarrow 8}, \delta_1^{\downarrow 8}] = \mathcal{C}(\Psi_0), \qquad \delta_i = \mathrm{ConvexUp}\big(8\,\delta_i^{\downarrow 8},\ \mathrm{softmax}(W_0)\big). \quad (9)$$

Reusing W_0 keeps the residual aligned to the motion boundaries of F at no parameter cost. The output layer of $\mathcal{C}$ is zero-initialized, so training starts from the linear model. Velocity is ill-defined under occlusion, so the residuals are gated before forming the velocities,

$$\alpha = \sigma(w_1 - w_2 U), \qquad m_i = F + \alpha\,\delta_i, \quad (10)$$

after which A, B follow from Eq. (3). The head is shared and applied a second time to Ψ_1, built from the already-available backward-pass state, giving A', B'. Neither pass depends on t, so both are evaluated once per frame pair. *Convex upsampling weights and layer specifications are given in Sec. A.4 and Sec. B of the supplementary.*

3.3. Integration with frame interpolation

Eq. (2) produces flow on the grid of I_0, so we transfer it to the target grid by softmax splatting [17] with negated displacement,

$$\tilde{G}_0 = \overrightarrow{\varpi}\big(-\Phi(t),\, \Phi(t),\, Z\big), \qquad \tilde{G}_1 = \overrightarrow{\varpi}\big(-\Phi'(t),\, \Phi'(t),\, Z'\big), \tag{11}$$

and fill unsplatted pixels, identified by their accumulated weight Ω_i, with a refinement network,

$$G_i = \tilde{G}_i + \mathrm{RefineNet}(\tilde{G}_0, \tilde{G}_1, \Omega_0, \Omega_1). \tag{12}$$

The per-pixel form of Eq. (11) is given in Sec. A.5 of the supplementary. Following flow-based VFI practice [5, 12, 17], the frames are backward warped, $I_{t\to i} = \overleftarrow{\varpi}(I_i, G_i)$, and fused by a mask and residual,

$$M, R = \mathrm{SynthNet}(I_{t\to 0}, I_{t\to 1}, G_0, G_1), \tag{13}$$

$$\hat{I}_t = M \odot I_{t\to 0} + (1-M) \odot I_{t\to 1} + R. \tag{14}$$

Pseudocode for inference and for a training iteration is given in Sec. C of the supplementary.

3.4. Loss

Reconstruction. We supervise reconstruction at the timesteps $\mathcal{T}$ available in each clip,

$$\mathcal{L}_{\text{rec}} = \sum_{t\in\mathcal{T}} \|\hat{I}_t - I_t\|_1 + \lambda_p \|\phi(\hat{I}_t) - \phi(I_t)\|_2^2, \tag{15}$$

with ϕ a fixed VGG feature extractor. Intermediate frames appear only as targets; the network receives no input beyond I_0 and I_1.

Velocity consistency. Eq. (2) and Eq. (4) describe one motion from opposite ends and must agree where they meet. Differentiating Eq. (4) at $t=1$ gives $\dot{\Phi}'(1) = -m_0'$, and a pixel at $\mathbf{x}$ arrives at $\mathbf{x} + F(\mathbf{x})$ on the grid of I_1, so

$$\mathcal{L}_{\text{vel}} = \big\|(1-\alpha) \odot \big(m_1 + \overleftarrow{\varpi}(m_0', F)\big)\big\|_1. \tag{16}$$

The derivation is given in Sec. A.2 of the supplementary. This term is a consistency constraint rather than a source of curvature, as $\delta \equiv 0$ satisfies it. The full objective is $\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_v \mathcal{L}_{\text{vel}}$.

4. Experiments

4.1. Setup

We train two models. Ours-V is trained on Vimeo90K-Septuplet [27], where motion is slow and close to linear. Ours-X is trained on X4K1000FPS [22], where it is fast and non-uniform. Each model is trained once and run on every benchmark, so both are tested in and out of distribution. Ours (linear) fixes $\delta_i \equiv 0$ and is otherwise identical.

We evaluate motion and frame quality separately. For motion we report PSNR$_f$ on normalized flows and EPE on unscaled flows, following [5]. On Vimeo-Septuplet-Flow the pseudo-ground-truth comes from FlowFormer [8]. On X-TEST-Flow we instead chain small-displacement flows through the native-rate frames, which is more reliable than one large-displacement estimate under this benchmark's fast motion. For frames we report PSNR and LPIPS [29] on X-TEST, SNU-FILM, and Vimeo90K-Septuplet.

The baselines are RIFE [9] and EMA-VFI [28] for linear motion, QVI [26] and EQVI [15] for four-frame quadratic motion, IQ-VFI [6] and GIMM-VFI [5] for two-frame nonlinear motion, and CT-VFI [1] in both its two-frame tracked setting (2^*) and its full-sequence spline setting (N). We run all of them with their released weights. The #F column gives the number of input frames each method needs.

Training uses AdamW, learning rate –, batch size –, – iterations on – GPUs at – crops. All supervised t within a pair share one motion pass.

Table 1. Intermediate motion, PSNR$_f$ ↑ / EPE↓. PSNR$_f$ is computed on normalized flows and EPE on unscaled flows. #F is the number of input frames. Ours-V is trained on Vimeo90K-Septuplet and Ours-X on X4K1000FPS; we evaluate both on all benchmarks.

		X-TEST-Flow		VSF
Method	#F	8×	16×	
Linear [10]	2			
QVI [26]	4			
EQVI [15]	4			
IQ-VFI [6]	2			
GIMM [5]	2			
CT-VFI [1]	2^*			
CT-VFI (spline) [1]	N			
Ours-V (linear)	2			
Ours-V	2			
Ours-X (linear)	2			
Ours-X	2			

4.2. Intermediate motion

Table 1 evaluates the trajectory before synthesis, which is where the parameterisation acts. Ours-X reduces EPE on X-TEST-Flow 16× by – over its linear variant and by – over GIMM, and it matches or beats the four-frame quadratic methods using two frames. CT-VFI in its base setting interpolates track positions linearly and performs accordingly. Its cubic-spline setting is stronger but needs the full sequence, so it is not a two-frame method; Ours-X reaches – EPE from two frames against its –. On Vimeo-Septuplet-Flow the margin is smaller, which is what we expect where motion is close to straight, but neither model falls below its linear variant.

4.3. Frame quality

The motion ordering carries through synthesis (Table 2). Ours-X gains – dB on X-TEST 16× over its linear variant and – dB over GIMM-VFI, with LPIPS improving by –. On SNU-FILM-arb the gap widens with motion magnitude, from – dB on Medium to – dB on Extreme. This is the trend the parameterisation predicts, since velocity residuals have nothing to correct when motion is already straight. On Vimeo-Septuplet, Ours-V stays within – dB of its linear variant while improving on the harder splits, so the added expressiveness costs nothing where it is unnecessary.

4.4. Where the velocity comes from

Flow is path independent: it records where a pixel starts and ends, not the route taken. A model given flow features alone should therefore drive δ_0, δ_1 to zero. Table 3 gates the inputs to CoeffHead at fixed head capacity, since a head too small to extract the signal would give a null we could not distinguish from its absence. The residual magnitudes, not the error metrics, are the result here. Flow-only collapses to $\|\delta_i\| = --$ and reproduces the linear baseline, as predicted. Appearance features raise it to – and cut EPE by –, and the blur descriptor adds a further –. The curvature comes from image content, not from flow.

We reuse RAFT's convex-upsampling mask ungated in the first row. It redistributes δ within each 8×8 block but cannot create magnitude where the low-resolution prediction is zero, so the test remains valid.

4.5. Number of supervised times

At a single t only $\beta_2(t)A + \beta_3(t)B$ is observable, so δ_0 and δ_1 are confounded: any pair reproducing that sum fits equally well. Table 4 varies the number of supervised times per pair at fixed gradient steps, so the comparison is one of supervision structure rather than budget. With one t the residuals are almost perfectly correlated ($\rho = --$) and EPE matches the linear baseline. At five times ρ falls to – and EPE improves by –.

This accounts for a prior negative result. IQ-VFI [6] reports no gain from an acceleration branch trained on reconstruction loss alone, on Vimeo triplets, which give a single t where the coefficients are unidentifiable by construction. The finding concerns identifiability, not the strength of photometric supervision.

4.6. Cost

The trajectory is parameter-free, so we fit the motion model once per keyframe pair and evaluate it K times, whereas implicit methods run theirs once per query time. Table 5 shows our motion cost flat in K while IQ-VFI and GIMM-VFI grow linearly. At $K = 32$ the motion model is $-\times$ cheaper.

Reversal and synthesis still run per t for every method, so total runtime grows with K. The amortisation applies to the trajectory model alone.

4.7. Qualitative results

Figure **??** compares interpolated frames under fast non-uniform motion. Figure **??** overlays the trajectory of a tracked point predicted by the linear variant, GIMM-VFI, and ours, against the ground-truth track from the dense sequence. Figure **??** maps $\|\delta_i\|$ spatially, showing where the model departs from constant velocity.

5. Conclusion

References

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Table 2. Frame interpolation, PSNR↑ / LPIPS↓. Notation as in Table 1.

		X-TEST		SNU-FILM-arb			VSF
Method	#F	8×	16×	Medium (4×)	Hard (8×)	Extreme (16×)	
RIFE [9]	2						
EMA-VFI [28]	2						
QVI [26]	4						
EQVI [15]	4						
IQ-VFI [6]	2						
GIMM-VFI [5]	2						
CT-VFI [1]	2*						
CT-VFI (spline) [1]	N						
Ours-V (linear)	2						
Ours-V	2						
Ours-X (linear)	2						
Ours-X	2						

Table 3. CoeffHead input ablation, Ours-X on X-TEST-Flow 16×. $\|\delta_i\|$ in pixels.

Inputs	PSNR$_f$ ↑	EPE↓	$\|\delta_0\|$	$\|\delta_1\|$
Flow only				
+ appearance $\mathcal{S}$				
+ blur $\mathcal{B}$				

Table 4. Number of supervised query times per frame pair, Ours-X on X-TEST-Flow 16×.

Times per pair	PSNR$_f$ ↑	EPE↓	$\|\delta_i\|$	$\rho(\delta_0, \delta_1)$
1 ($t = 1/2$)				
3				
5				

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Table 5. Runtime (ms) versus number of interpolated frames, —— $\times$ —— input on a single –. Motion is the trajectory model alone; total includes flow reversal and synthesis.

Method	Params (M)	K=1	K=8	K=16	K=32	Slope (ms/frame)
IQ-VFI [6]						
GIMM-VFI [5]						
CT-VFI [1]						
Ours (motion)						
Ours (total)						

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